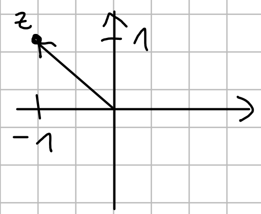
Lösungsvorschlag Probe-Prüfung Lineare Algebra

1. $z = i - 1$

a)

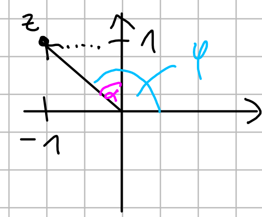


b) $r = |z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

$\alpha = \arctan\left(\frac{1}{1}\right) = 45° \hat{=} \frac{\pi}{4}$

$\varphi = 90° + \alpha = 135° \hat{=} \frac{3\pi}{4}$

$\Rightarrow \underline{z = \sqrt{2}\, e^{i\frac{3\pi}{4}}}$



c) $z^{-1} = \frac{1}{z \cdot \bar{z}}\, \bar{z} = \frac{1}{(-1)^2 + 1^2} (-i - 1) = \frac{1}{2} (-i - 1)$

<u>ODER</u>: $z^{-1} = \frac{1}{\sqrt{2}\, e^{i\frac{3\pi}{4}}} = \frac{1}{\sqrt{2}}\, e^{-i\frac{3\pi}{4}}$

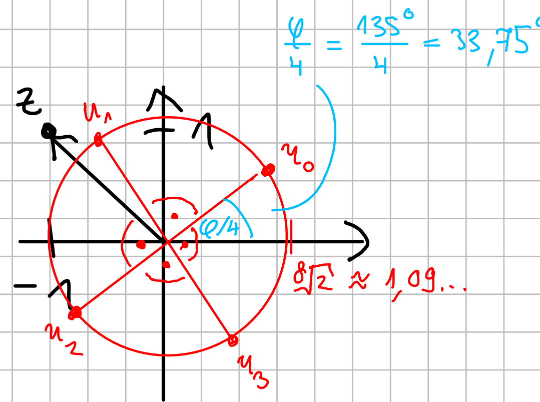
d) $u^4 = z = \sqrt{2}\, e^{i\frac{3\pi}{4}}$ besitzt Lösungen:

$u_0 = \sqrt[4]{\sqrt{2}}\; e^{i\frac{3\pi/4}{4}}$

$u_1 = \sqrt[4]{\sqrt{2}}\; e^{i\left(\frac{3\pi/4}{4} + 1 \cdot \frac{2\pi}{4}\right)}$

$u_2 = \sqrt[4]{\sqrt{2}}\; e^{i\left(\frac{3\pi/4}{4} + 2 \cdot \frac{2\pi}{4}\right)}$

$u_3 = \sqrt[4]{\sqrt{2}}\; e^{i\left(\frac{3\pi/4}{4} + 3 \cdot \frac{2\pi}{4}\right)}$



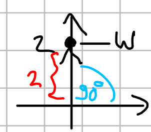
<u>ODER</u>: $u_k = \sqrt[4]{\sqrt{2}}\; e^{i\left(\frac{3\pi/4}{4} + k \cdot \frac{2\pi}{4}\right)}$, $k = 0, 1, 2, 3$

2. $w = 2i$

a) $\left(\frac{1}{2}w\right)^{1024} = i^{1024} = (\underbrace{i^2}_{-1})^{512} = (-1)^{512} = 1.$

b) $w^{-1} = \frac{1}{w \cdot \bar{w}} \cdot \bar{w} = \frac{1}{0^2+2^2}(-2i) = -\frac{1}{2}i$

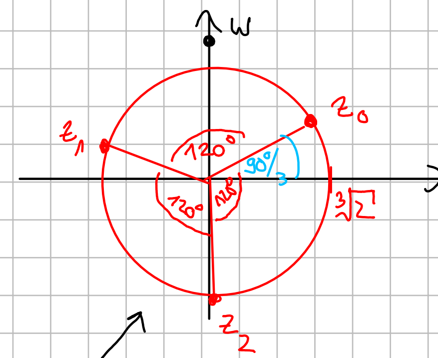
c) $z^3 = w = 2i = 2e^{i\frac{\pi}{2}}$



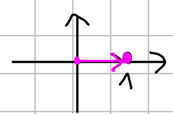
$z_0 = \sqrt[3]{2}\; e^{i\frac{\pi/2}{3}}$

$z_1 = \sqrt[3]{2}\; e^{i\left(\frac{\pi/2}{3} + 1\cdot\frac{2\pi}{3}\right)}$

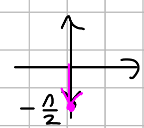
$z_2 = \sqrt[3]{2}\; e^{i\left(\frac{\pi/2}{3} + 2\cdot\frac{2\pi}{3}\right)}$



d) a)



b)



c)

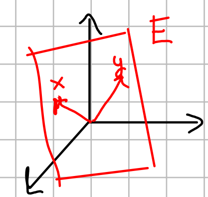
3. $(A|b) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 2 & 2a & -1 \end{array}\right) \sim_{II - 2\cdot I} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -2 & 2a-6 & -3 \end{array}\right)$

unabhängig von a immer lösbar & da x_3 frei immer ∞-viele Lösungen,

d.h. $\forall a \in \mathbb{R}$: LGS besitzt ∞-viele Lösungen.

4.

a) $E: \quad \lambda x + \mu y = \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix}$



b) $n = x \times y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix} = \begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix}$

<u>ODER</u>: $n = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ mit $n \cdot x = 0 \quad \& \quad n \cdot y = 0$

$$n \cdot x = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = n_1 + 2n_2 + 3n_3 \overset{!}{=} 0$$

$$n \cdot y = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix} = -2n_1 + 6n_2 + 3n_3 \overset{!}{=} 0$$

LGS: $\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ -2 & 6 & 3 & 0 \end{array}\right) \sim_{\mathrm{II} + 2 \cdot \mathrm{I}} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 10 & 9 & 0 \end{array}\right)$

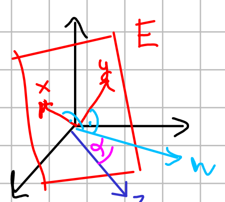
$n_3 = \alpha$ frei!

$\mathrm{II}: 10 n_2 + 9\alpha = 0 \quad \Rightarrow \quad n_2 = -\frac{9}{10}\alpha$

$\mathrm{I}: n_1 + 2 \cdot \left(-\frac{9}{10}\alpha\right) + 3\alpha = 0 \quad \Rightarrow \quad n_1 = -\frac{12}{10}\alpha$

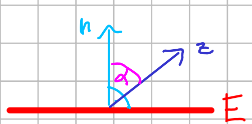
z. B. $n_3 = \alpha = 10$ liefert $n = \begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix}$.

c)



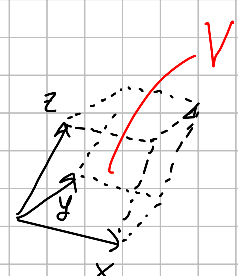
$$\alpha = \sphericalangle(z, n) = \arccos\left(\frac{z \cdot n}{|z| \cdot |n|}\right) = \arccos\left(\frac{\begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}}{\sqrt{12^2+9^2+10^2} \cdot \sqrt{1^2+2^2+4^2}}\right)$$

$$= \arccos\left(\frac{34}{5\sqrt{13} \cdot \sqrt{21}}\right) \approx 65{,}7^\circ$$



$$\sphericalangle(z, E) = 90^\circ - 65{,}7^\circ = 24{,}3^\circ.$$

d)



$$V = \left|\det(x|y|z)\right| = \left|\det\begin{pmatrix} 1 & -2 & -1 \\ 2 & 6 & 2 \\ 3 & 3 & 4 \end{pmatrix}\right| =$$

$$= \left| 24 + (-12) + (-6) - (-18) - 6 - (-16) \right|$$

$$= 34$$

(<u>ODER</u>: $V = \left|(x \times y) \cdot z\right| = \left|n \cdot z\right| = 34$ siehe c))

5. a)

$$x_2 = \frac{\begin{vmatrix} 1 & 1 & 3 \\ 0 & -1 & 2 \\ 0 & 2 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}} \overset{\text{Sarrus}}{=} \frac{-1-4}{1} = -5$$

b) EW:

$$\chi_A(\lambda) = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3, \text{ d.h. } \lambda_{1,2,3} = 1.$$

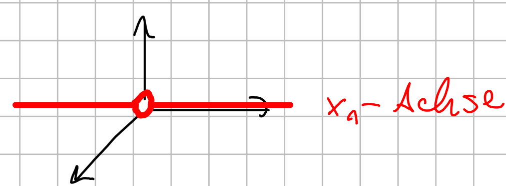
EV: $\lambda = 1$:

$$(A - \lambda E_3 | 0) = \left(\begin{array}{ccc|c} 1-1 & 2 & 3 & 0 \\ 0 & 1-1 & 2 & 0 \\ 0 & 0 & 1-1 & 0 \end{array}\right) = \left(\begin{array}{ccc|c} 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$x_1 = \alpha$ frei

II: $2x_3 = 0 \Rightarrow x_3 = 0$, I: $2x_2 = 0 \Rightarrow x_2 = 0$.

$$\Rightarrow \quad x = \begin{pmatrix} \alpha \\ 0 \\ 0 \end{pmatrix} = \underline{\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}$$



6. $B = \begin{pmatrix} 1 & 2b \\ 3b & 6 \end{pmatrix}$

a) B invertierbar $\Leftrightarrow$ $\det(B) \neq 0$

$\det(B) = 6 - 6b^2 = 6(1-b^2)$, also falls $1 - b^2 \neq 0 \Leftrightarrow \underline{b \neq \pm 1}$

Dann ist $B^{-1} = \frac{1}{6(1-b^2)} \begin{pmatrix} 6 & -2b \\ -3b & 1 \end{pmatrix}$.

b) B besitzt EW $\Leftrightarrow$ $\chi_B(\lambda)$ besitzt NST.

$\chi_B(\lambda) = \begin{vmatrix} 1-\lambda & 2b \\ 3b & 6-\lambda \end{vmatrix} = \underbrace{(1-\lambda)(6-\lambda)}_{\lambda^2 - 7\lambda + 6} - 6b^2 = 1 \cdot \lambda^2 - 7\lambda + (6 - 6b^2)$

besitzt NST, falls die Diskriminante $(-7)^2 - 4 \cdot 1 \cdot (6 - 6b^2) \geq 0$

$\Leftrightarrow \quad 49 \geq 24(1-b^2)$

$\Leftrightarrow \quad \frac{49}{24} \geq 1 - b^2$

$\Leftrightarrow \quad \underbrace{\frac{49}{24} - 1}_{\frac{25}{24}} \geq -b^2$

$\overset{\cdot(-1)}{\Leftrightarrow} \quad -\frac{25}{24} \leq \underbrace{b^2}_{\geq 0}$ $\quad \forall b \in \mathbb{R}$ erfüllt!

d.h. $\forall b \in \mathbb{R}$: B besitzt EW.

Link von Seite j auf Seite i an Matrixpos. i,j.

7.
$$A = \left(\frac{L_{ji}}{n_j}\right) = \left(\begin{array}{c|c|c} \frac{L_{11}}{n_1} & \frac{L_{21}}{n_2} & \frac{L_{31}}{n_3} \\ \frac{L_{12}}{n_1} & \frac{L_{22}}{n_2} & \frac{L_{32}}{n_3} \\ \frac{L_{13}}{n_1} & \frac{L_{23}}{n_2} & \frac{L_{33}}{n_3} \end{array}\right) = \left(\begin{array}{c|c|c} \frac{0}{1} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{1} & \frac{0}{2} & \frac{1}{2} \\ \frac{0}{1} & \frac{1}{2} & \frac{0}{2} \end{array}\right) = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{pmatrix}$$

Anzahl Links auf Seite j

Seite $j = 1 \mid 2 \mid 3$ Seite $j = 1 \mid 2 \mid 3$

Berechne EV zum EW $\lambda = 1$:

$$(A - 1 \cdot E_3 \mid 0) = \left(\begin{array}{ccc|c} 0-1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 0-1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0-1 & 0 \end{array}\right) \sim_{II+I} \left(\begin{array}{ccc|c} -1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & -1 & 0 \end{array}\right)$$

$$\sim_{III+II} \left(\begin{array}{ccc|c} -1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \begin{array}{l} -x_1 + \frac{1}{2}2\alpha + \frac{1}{2}\alpha = 0 \Rightarrow x_1 = \frac{3}{2}\alpha \\ -\frac{1}{2}x_2 + \alpha = 0 \Rightarrow x_2 = 2\alpha \end{array}$$

$x_3 = \alpha$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \alpha \begin{pmatrix} \frac{3}{2} \\ 2 \\ 1 \end{pmatrix}.$$ Für $\alpha = 1$ ergibt sich:

1. Seite 2 hat größtes Gewicht $x_2 = 2$
2. Seite 1 hat 2. größtes Gewicht $x_1 = 1{,}5$
3. Seite 3 hat 3. größtes Gewicht $x_3 = 1$